

Diagnosing Parkinson's By Applying Graph Neural Networks to MRI

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Abstract

Parkinson's Disease (PD) is a progressive neurodegenerative disorder and the second most common neurodegenerative disease in the elderly population. While it is characterized by degradation of dopaminergic neurons in the Substantia nigra, it is primarily diagnosed by motor symptoms and ruling out contradictory symptoms; this method is fairly error-prone, especially for early diagnosis. Machine learning methods have been applied to medical imaging to diagnose Parkinson's; however, one biomarker not yet explored is regional cortical thinning, which has certain advantages over past methodologies, which focused on volumetrics. By extracting cortical surfaces from MRI, converting the geometric mesh to a graph, and applying Graph Neural Networks, we may improve early-stage discrimination. These models were compared against a logistic regression baseline trained on clinical evaluation data, as well as a combined multimodal approach integrating both models. The GNN-based models alone demonstrated limited predictive performance but showed relative sensitivity to intermediate disease stages. Logistic regression achieved strong overall classification performance, while the combined model improved class balance, particularly for prodromal cases. These results suggest that graph-based structural representations capture complementary information related to early disease progression but are insufficient as standalone predictors. This multimodal approach may support more informed clinical assessment and has potential implications for earlier detection and improved disease management, where earlier diagnosis is associated with improved long-term outcomes.

I. Introduction

Parkinson's Disease affects over 1.1 million individuals within the United States [1]. 10-year mortality amongst individuals with PD is more than double of those without, at 47.9% versus 20.3% [2], and primarily affects those in the older age bracket, being the second-most common neurodegenerative disorder in the elderly [3]. While PD is typically characterized by degeneration of dopaminergic (dopamine-producing) neurons in the *substantia nigra*, a region of the midbrain [3], PD is *diagnosed* based on motor features and absence of contradictory symptoms, as this degradation is difficult to determine pre-mortem [4]. This method is prone to error; Rizzo et al. found a diagnosis rate of 80.6% [5], and Beach & Adler found an error rate of 58% for early diagnosis [6]. Thus, researchers are looking into other biomarkers, such as α -synuclein, to define and diagnose Parkinson's Disease [3]. One potential biomarker is regional thinning of the cortical surfaces, which was found in individuals with cognitive impairment but before onset of dementia, as Zhang et al. found [7].

A way to determine the extent of cortical thinning is to use Magnetic Resonance Imaging [8]. Specifically, T1-Weighted MRI was used in the study by Zhou et al. to determine the extent of Cortical Thinning [8]. T1-Weighted MRI provides high-quality anatomical images, allowing for extraction of detailed information on cortical surfaces [9]. One way to evaluate cortical thickness is to extract surfaces as geometric meshes via Freesurfer's recon-all pipeline, which contains information on the thickness, sulcal depth and other features at each point [10].

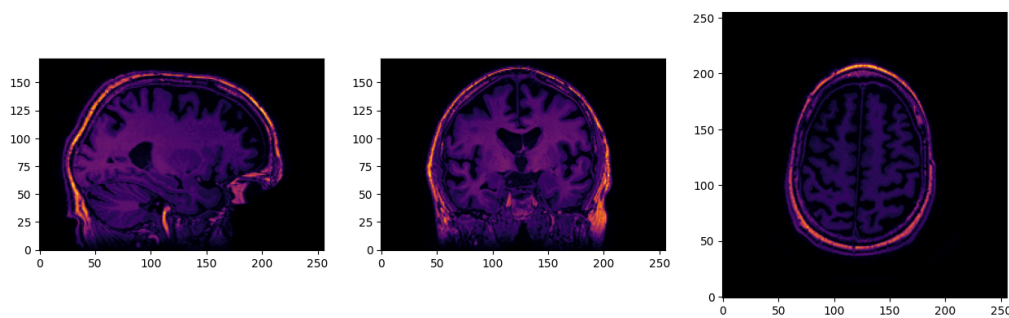
By converting these meshes to graphs, graph neural networks could be applied to extract information related to regional cortical thinning and possibly determine the degree of Parkinson's disease, allowing for earlier diagnosis. Graph neural networks for graph-level classification were able to achieve accuracy of 92.4% when applied to retinal imaging to determine degree of diabetic retinopathy [11]. More reliable early diagnoses could provide meaningful benefits, including improved symptom management and clinical decision-making. Parkinson's-specific treatments such as deep brain stimulation [12] and dopamine-mimicking agents [13] may help mitigate symptoms, although they do not currently alter disease progression. Additionally, younger age at diagnosis is associated with longer survival time [14].

II. Background

Past approaches applying machine learning techniques to diagnose Parkinson's have applied 3d CNNs to the volumes derived from MRI directly [15], or to images derived from DaTSCAN [16], which measures dopaminergic connection via injected radioactive tracer that binds to dopamine transporters [17]. Both of these approaches

have issues and advantages relative to the graph-based solution. High accuracy was observed in past studies, e.g. Chakraborty et al. attained 95.3% accuracy on the best testing fold [15]; however, 3d CNN is more computationally intensive due to having to convolve over the entire 3d volume rather than over the cortical surfaces. Chakraborty et al. only applied the CNN to 406 scans, which is relatively limited, and only differentiated between PD and control individuals [15]. Mallik et al. attained even higher accuracy, at >99% [16], but DaTSCAN has certain inherent limitations. It is significantly more expensive and time-consuming in comparison to MRI, which is comparatively more accessible and easier for most patients to access [18]. Thus, Graph NNs applied to cortical surfaces were identified as a promising research area.

MRI scans were sourced from the Parkinson's Progression Marker's Initiative, which collates MRI scans and disease status from subjects across many different studies [19]. Specifically, subjects are divided into control, Parkinson's Disease (**PD**), or prodromal, which refers to presence of mild Parkinsonism that has not been formally diagnosed as PD yet.



FreeSurfer's recon-all pipeline was applied to extract pial surfaces from the MRI scans; the pipeline additionally conforms the MRI scan to an atlas, which is useful for ensuring similar dimensions and resolution across surface graphs. It then segments the volume into certain regions utilizing the atlas, before determining white-gray matter boundary and extracting geometric mesh [10].

III. Applications:

This graph-based methodology can be extended in several meaningful ways when applied to cortical surface reconstructions generated by the FreeSurfer recon-all pipeline. After preprocessing structural MRI data into geometric meshes, the surfaces can be converted into graphs where nodes correspond to points and edges indicate spatial adjacency. This transformation enables the use of our graph-based model. This could potentially result in diagnoses years earlier, as early symptoms or prodromal state has been documented to appear 4-10 years before formal diagnosis of PD [20].

Clinically, these computational outputs are not intended to replace neurological expertise but to complement it. Graph-based predictions could be integrated alongside clinician assessments, patient history, and established imaging biomarkers (such as DaT-SPECT or volumetric MRI measures) to improve diagnostic confidence, particularly in early or ambiguous cases. Over time, this could support more personalized monitoring of disease progression and response to therapy, especially when combined with longitudinal MRI datasets.

IV. Methods:

T1-weighted MRI scans were sourced from the Parkinson's Progression Markers Institute (**PPMI**), with individuals from Control, PD and prodromal groups considered. 1772 scans were downloaded initially, with 339 control, 926 prodromal and 504 PD instances. These files were initially in the DiCOM format, and needed to be converted to NifTI, which was done using the dcm2niix tool [21]. Afterwards, scans were uploaded to the Open Science Grid (**OSG**), which provides large-scale distributed computing, allowing for efficient parallel processing [22]. The FreeSurfer (v7.1.4) recon-all pipeline was applied to each scan [10], with the extraneous information (e.g. non-surface information) being deleted to reduce memory load. Around 155 scans failed while extracting on the OSG, and we proceeded with a slightly reduced dataset, with 455 PD, 845 prodromal, and 314 control instances. 20% of the dataset was set aside as the testing set.

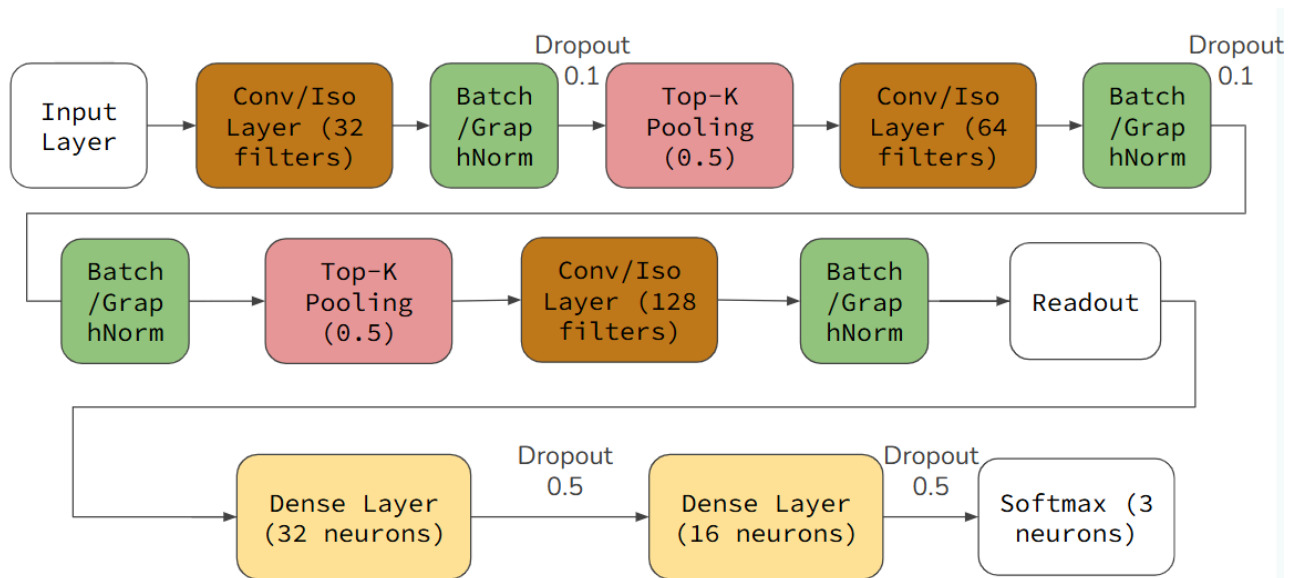
The resulting surface information was then converted into graphs by considering each point on the geometric mesh as a node, and connecting to the other points within the triangular face with edges. The features considered were the geometric coordinates (x,y,z) in millimeters, the thickness, the sulcal depth, the area, and the curvature of the cortical surface at said point.

Afterwards, data augmentation techniques were applied to create new instances and equalize the classes. Random noise was applied to the coordinates, to simulate slightly shifting the points, with noise being drawn from a normal distribution with mean 0 and standard deviation 0.05. Edges were randomly rewired (e.g. randomly cutting and adding edges between different nodes in such a way as to maintain the degree of the original node), with 5% of edges being rewired. The coordinates also had a small, random rotation and displacement applied, with rotation applied along both X and Z axes; the angle of rotation was drawn from a uniform distribution between -5 and 5 radians. Finally, the left and right hemispheres were combined with an offset being applied to the edge indices to allow them to be represented as a contiguous graph for each instance.

After data augmentation, the dataset contained 2792 instances, with 910 PD instances, 845 prodromal instances, and 942 control instances. Subsequently, the architecture of the graph neural network was determined. Two different architectures were considered, one based on Graph-Convolutional Networks (GCN) and one based on Graph-Isomorphic Networks (GIN). Graph-Convolutional Networks sum up features of adjacent nodes before normalizing by degree and then applying a learnable weight matrix [23]. Graph-Isomorphic networks are similar, except after the summing of neighboring features the features of each node are passed through a small, typically 2-layer, MLP, which allows greater expressiveness of each node and increased aptitude for graph-level classification [24].

Before pooling, batch or graph normalization was applied; batch normalization normalizes features across each minibatch to have mean 0 and standard deviation 1, whereas graph normalization normalizes features across each graph. Pooling with graph neural networks involves a few different methods; within this study, top-K pooling was used. Within top-K pooling, each node is given a score wherein the inner product is taken between the feature vector and a trainable weight vector before normalizing by magnitude of weight vector. Then, the top-K nodes are retained with the rest being discarded; these retained nodes' features are scaled by their scores passed through a tanh activation function [25].

The architecture used is represented in the below diagram:



For isomorphic layers, a 2-layer MLP was used with ReLU as the activation function. For readout, the node features directly after each pooling layer and of the final graph had the max and mean extracted and concatenated, and each of these representations were also concatenated to form the graph-level embedding fed into the densely-connected layers. LeakyReLU was used as the activation function, applied after normalization in the convolutional blocks as well as within the densely-connected layers.

The learning rate used was specifically 0.002, with AdamW used as the optimizer and categorical cross entropy as the loss function. For AdamW, β_1 , β_2 , and ϵ were set to 0.9, 0.999 and $1e-8$. Several different architectures were attempted, for experimentation, as described in Table 1.

Table 1: Models Attempted

Model 1	Model 2	Model 3
GCN & Batch normalization	GIN & Graph normalization	GCN & Graph Normalization w/ LogReg Model

Lazy-loading with gradient accumulation was used to avoid exceeding allowable memory and simulate larger batch sizes. Actual batches of 10 were accumulated over 2 iterations, leading to an effective batch size of 20.

Additionally, a multinomial logistic regression model with L2 regularization ($C=1.0$) was trained using the LBFGS optimizer on records of patient evaluations. These evaluations, both cognitive and motor, were also sourced from the PPMI; the records were all numeric. Specifically, the MDS-UPDRS Parts I, II and IV were merged, and missing values were imputed using Multiple Imputations by Chained Equations, which preserves relationships between features by predicting missing values using an estimator [26]. In our case, MICE was fit on the training set for 10 iterations with Bayesian Ridge Regression as the estimator. After MICE was run, a time restriction was applied, limiting each patient to their first 5 visits, to prioritize early diagnosis. If a patient's total visits were less than 5, all of their visits would be considered.

The resulting probabilities predicted by both the Logistic Regression model and the GCN model with graph normalization were fed into a densely-connected network, with the architecture 6 inputs, 9 hidden neurons, 5 hidden neurons and 3 outputs. This technique of combining models can lead to better accuracy than using either alone [27]. For the combined model, for each given instance of a scan, the average of the patient's first five visits would be used as input for the logistic regression model, as the augmented scans did not have an associated visit. Thus, various graph neural networks were applied to the augmented dataset of cortical surfaces extracted from MRI scans to determine if this method is able to accurately classify different degrees of Parkinson's disease.

V. Results:

Model 1:

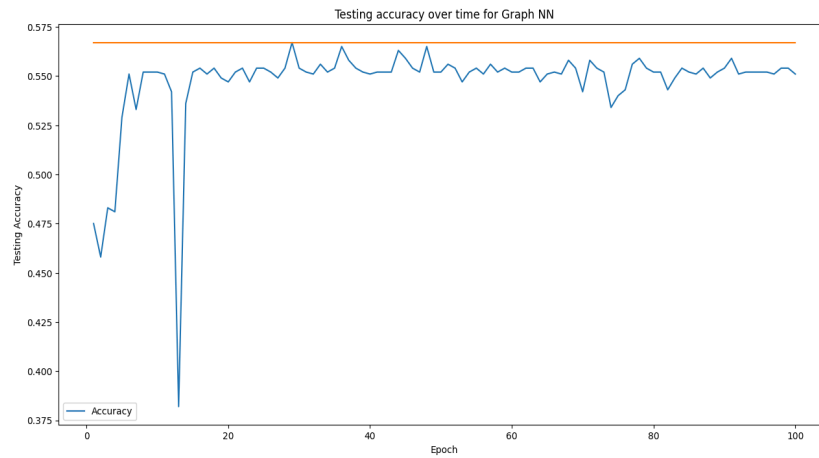


Figure 1: Testing Accuracy for Model 1

The model achieved 0.5671 testing accuracy, 0.799 recall on control, and 0.883 recall on Prodromal, as well as 0.5235 training accuracy. The confusion matrix is displayed below.

Table 2: Model 1 Confusion Matrix

Predicted→ Actual ↓	Control	Prodromal	PD
Control	151	38	0
Prodromal	22	166	0
PD	93	89	0

Model 2:

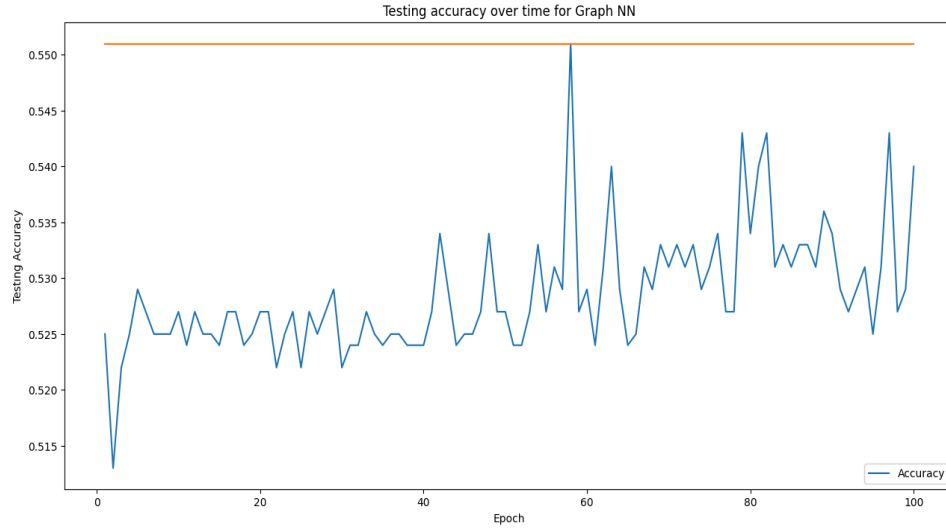


Figure 2: Testing Accuracy for Model 2

The model achieved 0.5510 testing accuracy, 0.672 recall on control, and 0.888 recall on prodromal, as well as 0.5481. The confusion matrix is displayed below:

Table 3: Model 2 Confusion Matrix

Predicted→ Actual ↓	Control	Prodromal	PD
Control	113	61	15
Prodromal	14	167	7
PD	57	97	28

Both graph-based models exhibit similar overall behavior, with low and nearly identical training and testing accuracy, at ~ 0.56 and ~ 0.55 testing accuracy for Model 1 and Model 2 respectively, indicating structured underfitting rather than successful feature learning. In both cases, the models are able to partially distinguish between control and prodromal subjects but fail to robustly learn a representation that separates all three classes. Figures 1 and 2 support this interpretation; the accuracy on the test set rises briefly, before oscillating for the remainder of training time, indicating that the model does not learn robust patterns in the representations.

A consistent failure mode across both models is the systematic misclassification of PD cases; Table 2 shows Model 1 failing to predict PD entirely, and Table 3 shows Model 2 predicting PD only infrequently. Importantly, this behavior occurs despite balanced class distributions, suggesting that the issue is not driven by class imbalance, but rather by insufficient feature separability in the learned graph representations. In other words, PD does not form a distinct cluster in the embedding space, and is instead absorbed into the decision regions of the other two classes.

While both models share this limitation, Model 2 shows a slightly more expressive decision boundary, as evidenced by Table 2, which displays its ability to occasionally identify PD cases. This suggests that the increased expressiveness of the GIN-based architecture may improve sensitivity to more complex class structures; however, this comes at the cost of increased misclassification across all classes, particularly between control and prodromal.

Overall, these results indicate that graph-based models capture some meaningful structural variation—particularly between control and prodromal—but are insufficient on their own to reliably distinguish PD, highlighting a limitation in representation quality rather than training convergence alone.

After training for 1000 iterations, the logistic regression model achieved an accuracy of 0.859 on the training set and 0.818 on the testing set. Confusion matrices are displayed below.

Training:

Table 4: Logistic Regression Train Confusion Matrix

Predicted→ Actual ↓	Control	Prodromal	PD
Control	843	64	36
Prodromal	137	1938	22
PD	9	183	1169

Testing:

Table 5: Logistic Regression Test Confusion Matrix

Predicted→ Actual ↓	Control	Prodromal	PD
Control	369	63	21
Prodromal	50	540	92
PD	8	44	344

The recall on testing is 0.815, 0.792, and 0.869 for control, prodromal, and PD respectively. The model slightly underpredicts Prodromal and has greater accuracy on PD.

The GCN and GraphNorm model alone achieved 0.5313 testing accuracy, 0.63 recall on control, and 0.846 recall on prodromal, as well as 0.5540 training accuracy. The confusion matrix is displayed below.

Table 6: GCN/GraphNorm Confusion Matrix

Predicted→ Actual ↓	Control	Prodromal	PD
Control	119	63	7
Prodromal	28	159	1
PD	82	81	19

The combined model achieved accuracy of 0.843 on the testing set. The confusion matrix is displayed below.

Table 7: Combined Final Confusion Matrix

Predicted→ Actual ↓	Control	Prodromal	PD
Control	154	23	12
Prodromal	11	162	15
PD	7	20	155

The logistic regression model trained on clinical evaluation features achieves strong and stable performance, with a testing accuracy of 0.818 and balanced class-wise recall (control: 0.815, prodromal: 0.792, PD: 0.869), as seen in Table 5. Tables 4 and 5 indicate relatively consistent classification across all three classes, with the highest performance observed for PD and slightly reduced sensitivity for the prodromal class. This suggests that clinical evaluation features are highly informative and provide strong linear separability, particularly for distinguishing advanced disease stages.

In contrast, the standalone graph-based model (GCN with GraphNorm) performs significantly worse, with a testing accuracy of 0.5313 and overall weaker class separation. Table 6 indicates that the GCN/GraphNorm model achieves reasonable recall for control (0.63) and prodromal (0.846); however, it struggles with PD classification, indicating that structural graph representations alone do not provide sufficient discriminative power for full disease staging. This reinforces earlier observations that graph-based features capture partial but incomplete information about disease progression, particularly favoring intermediate-stage detection while failing to robustly isolate PD.

The combined model, which integrates graph-based representations with logistic regression, achieves the best overall balance with a testing accuracy of 0.843. Table 7 shows improved uniformity across classes, with recall values of 0.815 (control), 0.862

(prodromal), and 0.851 (PD). Notably, this model improves prodromal detection compared to logistic regression alone, while maintaining strong PD performance.

This pattern suggests that the two feature modalities provide complementary information: clinical evaluation features are primarily responsible for strong separation of clearly defined classes (control and PD), while graph-based features contribute additional sensitivity to intermediate disease states such as prodromal. However, the improvement is not uniform across all classes, as the inclusion of graph features slightly redistributes decision boundaries rather than uniformly increasing performance. Overall, these results indicate that multimodal integration improves classification balance, particularly for the prodromal class, but that clinical features remain the dominant source of predictive power. Graph-based features provide secondary, complementary structure-related information that is most useful for capturing transitional disease stages rather than definitive class separation.

VI. Limitations:

One major limitation of this work is the lack of access to GPU resources on the TJHSST cluster. As a result, all model training had to be performed on CPU, which significantly increased training time and limited the number of experiments that could be feasibly run. This constraint restricted both hyperparameter tuning and the ability to explore larger or more complex architectures.

Another key limitation is the relatively small size of the neural network used in this project. The model contains approximately 26,000 parameters, which is substantially smaller than architectures used in prior related work. For comparison, studies such as [14] employ well-established deep convolutional neural networks including VGG-16, DenseNet, and InceptionV3, which range from roughly 8 million to 138 million parameters. Similarly, the custom CNN architecture in [8] contains approximately 41 million parameters.

Because of this difference in scale, the model in this work may have limited representational capacity compared to larger architectures. While neural networks with sufficient width can approximate a wide range of functions in theory (as established by the Universal Approximation Theorem), in practice the efficiency of this approximation strongly depends on network depth and parameter count. Shallower networks often require disproportionately more training data or neurons to represent the same class of functions that larger or deeper networks can express compactly, which can lead to reduced performance when modeling highly complex or hierarchical structure in data.

Research in deep learning theory has shown that depth in particular plays a crucial role in expressivity: deeper networks can represent certain functions exponentially more efficiently than shallow or narrow ones. For example, [28] demonstrates that there exist functions that can be represented with polynomial-size

deep networks but would require exponentially larger shallow networks to approximate with comparable accuracy. Similarly, [29] shows that certain radial functions can be efficiently represented with depth-3 networks but not with shallow ones unless they grow exponentially in size. Empirical analyses such as [30] further support the idea that deeper, higher-capacity networks learn richer hierarchical feature representations, which are often necessary for capturing complex structure in real-world data.

Given that the model in this work contains only ~26,000 parameters—several orders of magnitude fewer than architectures such as VGG-16, DenseNet, or InceptionV3—it is likely constrained in its ability to represent similarly rich hierarchical features. As a result, the model is likely limited in its ability to capture more intricate relationships present in the surface graphs compared to these larger, more expressive architectures.

VII. Conclusion:

This study explored regional cortical thinning as a biomarker of Parkinson's disease (PD), wherein cortical surface reconstructions are extracted from structural MRI data and transformed from geometric meshes into graph representations. Using the FreeSurfer recon-all pipeline, T1-weighted MRI data is transformed into cortical surface graphs defined by vertices and spatial adjacency. This transformation enables graph neural networks to operate directly on cortical geometry, allowing them to model spatially localized patterns of cortical thickness variation associated with neurodegeneration.

Within this framework, graph neural networks were implemented using GCN and GIN architectures, operating on these cortical graph representations, while a separate baseline logistic regression model was trained on engineered clinical evaluation features. A multimodal model was then constructed by combining graph-based embeddings with clinical feature-based predictions to evaluate whether structural and clinical information provide complementary signals for disease classification.

Overall, the results indicate that clinical evaluation features provide the strongest standalone predictive signal. The logistic regression model achieved the highest and most stable performance, with a testing accuracy of 0.818 and balanced class-wise recall across control (0.815), prodromal (0.792), and PD (0.869), suggesting that these features are highly informative and largely linearly separable.

In contrast, the graph-based models achieved substantially lower performance (testing accuracy ~0.53–0.57), exhibiting underfitting and limited representational capacity. In particular, both GCN and GIN models struggled to reliably distinguish PD cases, with samples frequently misclassified as control or prodromal. This suggests that PD does not form a clearly separable structure in the learned embedding space, and that graph representations alone are insufficient for robust disease staging in this setting.

Despite these limitations, graph-based representations were not entirely uninformative, as they captured partial structure useful for distinguishing control from prodromal subjects, indicating sensitivity to early disease variation. The multimodal model combining graph embeddings with logistic regression achieved the best overall balance, reaching a testing accuracy of 0.843. This improvement suggests that graph-derived features provide complementary structural information to clinical variables, particularly for intermediate disease stages, while clinical features remain the dominant source of class separability.

Overall, these findings suggest that while graph-based learning alone is insufficient for Parkinson's disease classification, it provides meaningful structural information that becomes most effective when integrated with clinical features, supporting its role as a complementary modality in multimodal disease modeling.

VIII. Future Work:

Several directions could further improve model performance and better leverage graph-based representations.

First, further tuning of the GCN/GIN architectures could help address the observed underfitting. This includes experimenting with increased model capacity, alternative aggregation functions, and improved regularization strategies such as dropout, residual connections, or normalization techniques. Given the limited expressiveness observed in the current models, architectural changes that enhance depth or stabilize training may improve class separability, particularly for PD cases.

Second, the feature construction and graph design process could be refined. The current results suggest that PD does not form a clearly separable structure in the learned embedding space, which may indicate that the input graph representation does not fully capture disease-specific relationships. Exploring alternative edge definitions, feature engineering strategies, or incorporating temporal or longitudinal relationships (if available) may improve representation quality.

Third, the multimodal approach could be extended beyond simple concatenation or linear combination. More advanced fusion strategies—such as attention-based fusion, learned weighting of modalities, or end-to-end joint optimization—may better exploit complementary information between graph-based and clinical features.

Finally, the performance gap between graph-based models and traditional machine learning methods suggests that additional experimentation with pretrained deep learning architectures on alternative representations (e.g., 3D CNNs such as VGG-style variants or Inception-based models) would be valuable. Comparing these against improved graph models would help determine whether structural graph information can ever match or exceed performance from image-based or tabular clinical representations.

IX. Materials:

This study utilized T1-weighted 3D structural MRI data and clinical evaluation data from the Parkinson's Progression Markers Initiative (**PPMI**) database. Clinical labels and features were derived from MDS-UPDRS Parts I, II, and IV, which capture non-motor experiences of daily living, motor experiences of daily living, and motor complications, respectively. These measures were used to define class labels for control, prodromal, and Parkinson's disease (PD) stages.

Structural MRI scans were processed using the FreeSurfer (v7.4.1) recon-all pipeline to generate cortical surface reconstructions. These surfaces were then used to construct graph representations for downstream learning tasks.

All data preprocessing, feature engineering, and analysis were implemented in Python using NumPy and Pandas for data manipulation and organization. Machine learning models were developed using scikit-learn for the logistic regression baseline and PyTorch Geometric for implementation and training of graph-based models, including Graph Convolutional Networks (**GCN**) and Graph Isomorphism Networks (**GIN**). A multimodal model was additionally constructed by combining graph-based embeddings with clinical features to evaluate complementary predictive performance across modalities.

All experiments were conducted on the TJHSST computing cluster, with training performed under CPU-only constraints due to the absence of GPU resources.

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APPENDIX:

I. CODE:

[Include under shared Google drive (you may include zip file)]

II.